

Quiz 3

Question 1:

A smartphone manufacturer gets replacement batteries from three suppliers, S1, S2, and S3.

- Supplier S1 supplies **40%** of the batteries.
- Supplier S2 supplies **35%**.
- Supplier S3 supplies **25%**.

Each supplier has a different defect rate:

- S1: **1.5%** of their batteries are defective.
- S2: **2.8%** are defective.
- S3: **5.0%** are defective.

A battery is randomly selected from the day's incoming stock and is found to be **defective**.

- (a) What is the probability that the defective battery came from supplier S1? [2]
(b) What is the probability it came from supplier S2? [2]
(c) If the manufacturer tests one battery and it's **non-defective**, what is the probability it came from S1? [3]

Question 2:

A bank wants to predict whether a **loan application will be approved** ("Yes") or rejected ("No") based on three features:

1. **Credit Score**: {High, Low}
2. **Employment**: {Salaried, Self-Employed}
3. **Marital Status**: {Married, Single}

The bank collected the following dataset of 10 applicants:

Applicant	Credit Score	Employment	Marital Status	Approved?
1	High	Salaried	Married	Yes
2	Low	Self-Employed	Single	No
3	High	Salaried	Single	Yes
4	Low	Salaried	Married	No
5	High	Self-Employed	Married	Yes
6	Low	Self-Employed	Married	No
7	High	Salaried	Married	Yes
8	Low	Salaried	Single	No
9	High	Self-Employed	Single	Yes
10	Low	Salaried	Married	No

Predict whether a **new applicant** with the following profile will be **approved** or **rejected**:

- Credit Score = **High**
- Employment = **Salaried**
- Marital Status = **Single**

Quiz 4

You need to choose a model (**Linear/Logistic Regression**) to predict whether the car is fuel-efficient based on two features: Engine Size (liters) and Car Weight (100 kg units).

Engine_Size(L)	Car_Weight(100kg)	Fuel Efficient (1=Yes, 0=No)
1.6	12	0
2	15	1
2.4	14	0

Consider all initial parameters as 0.05 and the learning rate as 0.02.

Show one iteration (including optimization) using the parameters given above.

Part B:

State and explain whether the following answers are true or false

1. In simple linear regression, if the gradient with respect to m is zero, the cost function has not reached a global minimum.
2. Logistic regression is equivalent to linear regression if you remove the sigmoid function.
3. In logistic regression, if the predicted probability for a positive class is 0.9 but the true label is 0, the loss contribution for this example is large.

Set C

Show expression for the forward propagation only

